

Name
i) Compare & shift (if not match) then Pattern

Eg ① a b c d e f g h — n
def m $O(m, n)$

③ a b c d a b e a b c d f if no match
a b c d f
if match i++, j++
j → 0 next

Drawback
Repeatedly checking

Worst case a a a a a a b
a a a b

KMP
Pattern: a b c d a b c
(left → right)
Prefix: a, ab, abc, abcd
Suffix: c, bc, abc, dabc

Prefix same as Suffix

P₁: $\begin{matrix} 1 & 2 & 3 & 4 \\ \underline{a} & \underline{b} & \underline{c} & \underline{d} & a & b & e & a & b & f \\ 0 & 0 & 0 & 0 & 1 & 2 & 0 & 1 & 2 & 0 \end{matrix}$

P₂: $\begin{matrix} a & b & c & d & e & a & b & f & a & b & c \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 & 1 & 2 & 3 \end{matrix}$

P₃: $\begin{matrix} a & a & b & c & a & d & a & a & b & c \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 2 & 0 & 0 \end{matrix}$

P₄: $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\ \underline{a} & \underline{a} & \underline{a} & \underline{a} & \underline{b} & \underline{a} & \underline{a} & \underline{c} & \underline{d} \\ 0 & 1 & 2 & 3 & 0 & 1 & 2 & 0 & 0 \end{matrix}$

KMP
String $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\ a & b & a & b & c & a & b & c & a & b & a & b & a & b & d \end{matrix}$ — n

Pattern
Table

a	b	a	b
0	0	1	2

 m

if $S[i] == P[j+1]$ *
j++, i++

if no match
move to index 2

if we unable to move j further
then move i

$O(m+n)$
Time Search